RoboHairSim: A Human-Machine Platform for Planning and Evaluating Robotic Haircutting

***Abstract*—Robotic haircutting is a typical human-machine system with both robots and human customers involved. It is still at an early stage, and there is no means of testing and comparing haircutting methods under repeatable conditions. Trials on real hair are hard to repeat, because every cut changes the hair irreversibly and a given hair state cannot be reproduced. This paper presents RoboHairSim, a simulation platform for planning and evaluating robotic haircutting. It models the head with its scalp regions, the hair with its flow direction and cutting state, the clipper and the robot arm in one scene. Clipper toolpaths are planned over each scalp region and turned into a continuous robot motion that satisfies the scene constraints, and a person can also drive the same robot by teleoperation. An effective-cutting criterion based on the guard comb decides whether a clipper pose cuts a hair, and the same criterion evaluates both the planned toolpaths and the executed motion. Layered coverage metrics thus attribute a coverage loss to planning or to execution. Experiments on one head compare seven toolpath combinations, three robot mounting configurations and two execution modes, and complete continuous full head haircuts and a teleoperated haircut of one region. They show that geometric coverage alone overestimates what a toolpath achieves, and that the losses added by robot execution depend on the mounting configuration and can be located on the scalp.**

***Index Terms*—Robotic haircutting, simulation, human–robot interaction, motion planning, coverage evaluation**

I. Introduction

Haircutting is a routine part of personal care. It is repetitive yet skilled labor that takes barbers a long training period [1]. It also remains hard to access for people with limited mobility, such as the elderly and patients in long-term care, and in places without professional barbers [2], [3]. Automating haircutting with robots could ease this labor demand and widen access, yet research on robotic haircutting remains at an early stage, with only a few scattered attempts [4]. Part of the difficulty lies in executing the task. The clipper has to move continuously close to the curved head and cover the hair to be cut, while avoiding sensitive areas such as the ears, eyes and face. Moreover, the result of a haircut is expressed in the hair itself, whose many easily deformed strands vary in length, density and growth direction between individuals [5].

The idea of automating haircutting first appeared in patents. One proposed a machine that cuts a predetermined style under a preset program [6], and another a robot with sensing and a user interface [7], but both remained at the level of design concepts. Futami *et al.* [8] used a camera-equipped mobile robot to assist users in cutting their own hair, the cutting itself being done by the person. Robotic studies of adjacent hair-care tasks, including head care [9], detangling [10], combing [11], [12] and styling [13], have advanced hair perception and contact manipulation but do not address cutting. In recent years, studies dedicated to robotic haircutting have gradually emerged. They examine the gap between theory and practice [14] and draw an analogy between haircutting and computer numerical control (CNC) machining [15]. Another study distinguishes the space swept by the tool from the region where cutting actually takes place [16]. A teleoperation baseline has further demonstrated robotic haircutting on a mannequin, with an operator guiding the clipper through visual feedback [17].

Even as autonomy improves, robotic haircutting is unlikely to take the barber or caregiver out of the loop. A person may teleoperate the robot, supervise an autonomous cut, or take over to finish or personalize it, for example by giving the sides a different length from the top. In each of these roles, the person's commands must obey the same safety constraints as those of a planner. The result must also be judged by the same notion of a successful cut. A tool for developing robotic haircutting should therefore serve both autonomous planning and human operation, and evaluate them in the same way.

In both cases, a cutting plan has to be tried and compared repeatedly before real hair is cut. Such trials are hard to conduct physically. Testing on people carries safety risks, while on a wig or mannequin head every cut changes the hair irreversibly. A given hair state therefore cannot be reproduced, and different methods, operators and robot configurations cannot be compared under the same conditions. Reviews of safety measures for haircutting close to the body [18] and of autonomous haircutting [19] likewise identify the lack of validation methods and reproducible benchmarks. Fields such as manipulation [20], navigation [21] and autonomous driving [22] meet this need with simulation environments, where methods are developed, tested and compared before deployment on real systems.

No simulation environment yet treats robotic haircutting as a whole, and existing tools each cover only part of it. In computer graphics, interactive hairstyling systems simulate cutting with virtual scissors [23]–[25], but the cut is decided by the user and judged visually, and no robot is involved. General-purpose robot simulators [26]–[28] model arm kinematics, collisions and reachability but contain no model of hair or cutting. Surface coverage planning [29], [30] and CNC toolpath generation [31] measure coverage by the region swept by the tool. In haircutting, however, the cut also depends on the distance of the tool from the scalp and on the feed direction relative to the hair growth. This dependence is what makes evaluation specific to haircutting. A clipper cuts with a guard comb that lifts the hair only when the clipper is pushed against the hair flow close to the scalp. A pass that sweeps over the hair in the wrong direction therefore leaves it uncut. Coverage measured by the swept region thus overestimates what a toolpath achieves. An evaluation has to model this effective cutting and apply it both to the planned toolpath and

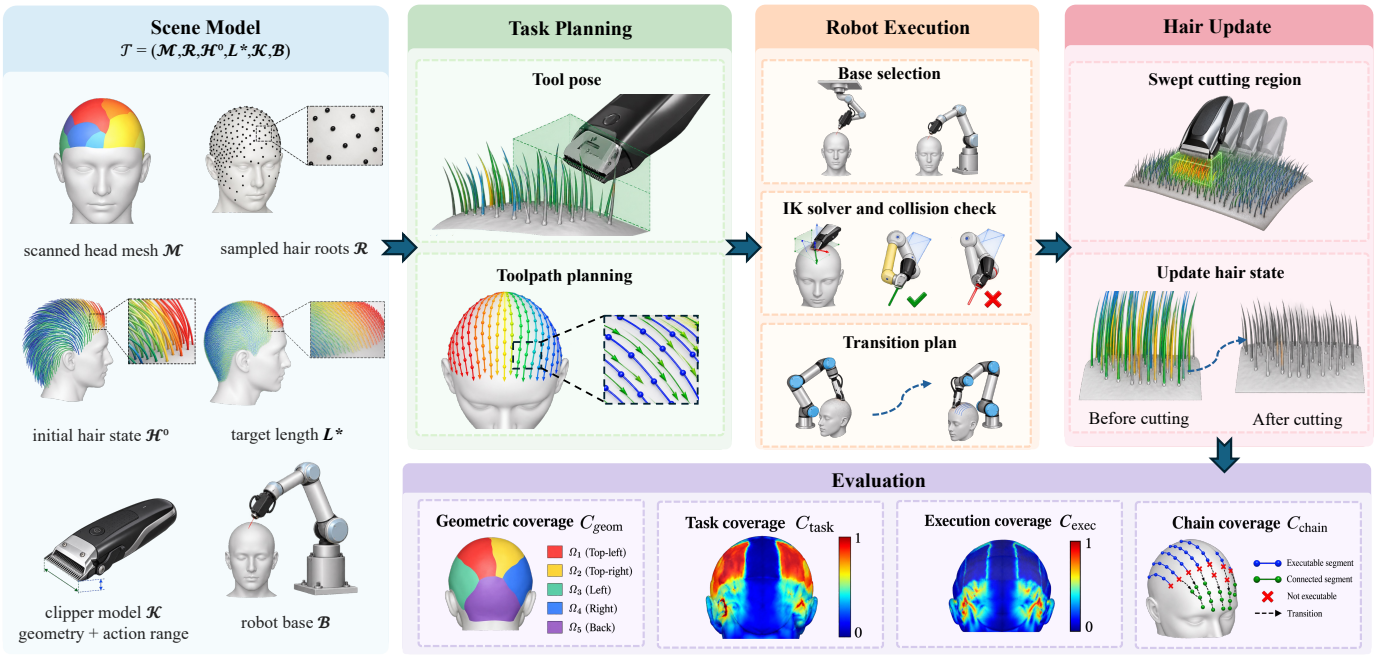


Fig. 1. Architecture of RoboHairSim. The platform integrates scene modeling, task planning and robot motion planning. A shared effective-cutting criterion governs the hair state update and the coverage evaluation.

to the executed robot motion. Only then can a shortfall be attributed to planning or to execution.

We therefore present RoboHairSim, a task-level simulation platform for planning and evaluating robotic haircutting (Fig. 1). It models the head, the hair, the clipper and the robot arm in one scene, where clipper toolpaths can be planned and executed by the arm or the robot can be driven by a person through teleoperation. In both modes, one effective-cutting criterion cuts the simulated hair and evaluates the planned and executed results. This paper makes the following contributions.

1) We develop RoboHairSim, to our knowledge the first simulation platform for robotic haircutting, supporting both autonomous planning and teleoperation in the same scene under the same constraints and cutting criterion.
2) We propose an effective-cutting criterion that applies to both planned toolpaths and robot execution, and define layered coverage metrics that attribute coverage losses to planning or execution.
3) We implement a reference planning method and conduct benchmark experiments on toolpath components, mounting configurations and execution modes.

The rest of this paper is organized as follows. Section II presents the scene model and the effective-cutting criterion. Section III describes task planning, robot motion planning, teleoperation and the coverage evaluation. Section IV reports the experiments, and Section V concludes the paper.

II. Haircutting Task and Scene Modeling

This section defines the haircutting task and models the scene in which it is carried out. The scene model describes the scalp regions and hair roots, the hair, and the clipper with its effective-cutting criterion, which decides whether a clipper pose cuts a hair, together with the scene constraints, which decide whether a clipper pose and an arm configuration are admissible. Section III then builds on these two models to plan the haircut and to evaluate its result.

A. Haircutting Scene Definition

A haircutting task is represented as

$$\mathcal{T} = (\mathcal{M}, \mathcal{R}, \mathcal{H}^0, \boldsymbol{L}^\star, \mathcal{K}, \mathcal{B}), \tag{1}$$

whose components are listed in Table I and whose geometry is expressed in one world frame. Roots and hairs are indexed by $i = 1, \ldots, N$, hair nodes by $k = 1, \ldots, K$ and cutting updates by t. The active node count $a_i(t)$, recording how far each hair has been cut, forms the task state, whereas the hair node positions, clipper pose and arm configuration together form the simulation state.

B. Scene Modeling

1) Scalp regions and roots: The scalp triangles of the head mesh $\mathcal{M} = (V, F)$ are divided into n_Ω treatment regions $\mathcal{D} = \{\Omega_m\}_{m=1}^{n_\Omega}$, while the remaining triangles belong to the face, ears, neck or hairless skin. The label $\Omega(\cdot)$ gives the region of a triangle or a root. Hair roots are sampled on the scalp with a probability proportional to the triangle area and a regional density weight, at uniformly distributed barycentric coordinates. The candidates are thinned toward the frontal hairline, removed within a band along the parting and kept only behind the front edge of the ears, and each new root must be at least δ_r away from the existing ones. The result is the root set $\mathcal{R} = \{(\boldsymbol{r}_i, \boldsymbol{n}_i^{\triangle}, \Omega(i))\}_{i=1}^{N}$ of root positions, unit normals of their triangles and regions. In this work the scalp has five regions, the top left and top right regions, the left and right sides and the back (Fig. 2).

TABLE I
KEY PARAMETERS OF THE HAIRCUTTING TASK.

Symbol	Meaning
$\mathcal{M} = (V, F)$	Labeled head mesh (vertices V, triangles F)
$\mathcal{D} = \{\Omega_m\}_{m=1}^{n_\Omega}$, $\Omega(\cdot)$	Scalp regions, region label
$\mathcal{R} = \{(\boldsymbol{r}_i, \boldsymbol{n}_i^{\triangle}, \Omega(i))\}_{i=1}^{N}$	Root position, triangle normal, region
$\mathcal{H}^0$	Initial hair state
$\boldsymbol{L}^\star = (L_i^\star)_{i=1}^N$	Target length of each hair
$\mathcal{K}$	Clipper geometry and action parameters
$\mathcal{B}$, n_q, T_{wb}	Robot arm and base, joint count, base pose
$\{\boldsymbol{g}_{i,k}^0\}_{k=1}^K$	Reference shape of the hair
$\ell_i(k)$, $a_i^\star$	Reference arc length, target node count
$\boldsymbol{f}_i$	Hair flow direction
$a_i(t)$, $\mathcal{U}(t)$	Active node count, set still to be cut
$\boldsymbol{g}_{i,k}(t)$	Hair node positions
T_{wc}, $\boldsymbol{q}$	Clipper TCP pose, arm configuration

Note. Given, derived and state quantities are listed from top to bottom.

2) Hair model: Each root generates one hair, which represents a tuft of strands around the root and is the unit of the coverage and length evaluation. Planning, execution and evaluation use only the root positions, the hair flow direction and the reference arc length of each hair, so any hair model that provides these quantities can be used.

a) Reference shape: The i-th hair is a polyline of K nodes with reference shape $\{\boldsymbol{g}_{i,k}^0\}_{k=1}^K$, generated by a procedural hairstyle model from a root node slightly above the root along the triangle normal. Its length is drawn within the range of its region, and each segment follows a weighted combination of the scalp normal, a regional hairstyle flow and gravity, so that the hair lifts near the root, follows the hairstyle in the middle and droops toward the tip. A middle parting divides the top into a left and a right region, and to imitate a crown whorl the hair of the two top regions lies forward in front of the highest point of the head and backward behind it, while the hair of the side and back regions points mainly downward, as shown in Fig. 2b.

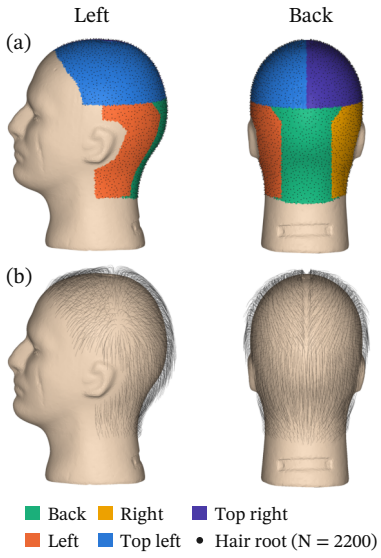


Fig. 2. Scalp regions and hair (left and back views). (a) The five scalp treatment regions (colors) and the hair roots (black dots). (b) Reference shape of the hair, with one polyline per hair.

b) Target node count: With $\ell_i(k)$ the length of the reference polyline from the root node to node k, the target length $L_i^\star$ is converted to the target node count, the first node at which this arc length reaches the target,

$$a_i^\star = \begin{cases} \min\{k : \ell_i(k) \geq L_i^\star\}, & \text{if this set is non-empty,} \\ K, & \text{otherwise,} \end{cases} \tag{2}$$

so the retained length exceeds the target by less than one node spacing. Nodes $1, \ldots, a_i(t)$ form the retained part of the hair, with $a_i(0) = K$, and the set of hairs still to be cut at step t is

$$\mathcal{U}(t) = \{i : a_i(t) > a_i^\star\}. \tag{3}$$

The goal of the haircutting task is to make $\mathcal{U}(t)$ empty.

c) Morphological evolution: The node positions $\boldsymbol{g}_{i,k}(t)$ are updated in the manner of position-based dynamics [32], with restoration toward the reference shape, gravity, damping, bending and segment length constraints and collision with the head, and the nodes removed by cutting fall as clippings. This evolution only displays the hair, since cutting and evaluation use the reference shape.

3) Hair flow direction: With $\boldsymbol{n}_i$ the smoothed scalp normal at the root node, the hair flow direction is the normalized projection onto the tangent plane of the displacement from the root node to the middle node $k_{\text{mid}} = \lceil K/2 \rceil$,

$$\boldsymbol{f}_i = \frac{(I - \boldsymbol{n}_i \boldsymbol{n}_i^\top)(\boldsymbol{g}_{i,k_{\text{mid}}}^0 - \boldsymbol{g}_{i,1}^0)}{\|(I - \boldsymbol{n}_i \boldsymbol{n}_i^\top)(\boldsymbol{g}_{i,k_{\text{mid}}}^0 - \boldsymbol{g}_{i,1}^0)\|}. \tag{4}$$

Since the clipper works close to the scalp, the direction is taken from the part of the hair near the root, which is not affected by the drooping tip.

4) Clipper and effective cutting: The tool center point (TCP) frame of the clipper lies at the center of the teeth face, with the x axis along the feed direction, the y axis along the blade width and the z axis pointing into the scalp. Its world pose is $T_{wc} = (R_{wc}, \boldsymbol{p}_{wc}) = T_{we} T_{ec}$, where T_{we} is the pose of the arm's end flange and T_{ec} the clipper mounting transform. The whole clipper shape, a point set $\mathcal{Q}(T_{wc})$ at pose T_{wc}, is used for the clearance constraint, whereas the cutting decision uses only the action range in front of the teeth.

In practice, the length left by a clipper is set by its guard comb. When the clipper is pushed against the hair flow close to the scalp, the comb lifts the hair and holds it upright between its teeth, and the blade cuts the part protruding beyond the comb. The cutting model follows this practice. A hair is cut directly to its target length when its root node lies in the action range in front of the teeth, the blade is close enough to the scalp and the clipper is pushed against the hair flow, without simulating blade–strand contact. The clipper pose thus decides only whether a hair is cut, and the retained length is set by the target. To evaluate a pose, the root node is first transformed to TCP coordinates as

$$\boldsymbol{r}_i^c = R_{wc}^\top (\boldsymbol{g}_{i,1}^0 - \boldsymbol{p}_{wc}) = (r_{i,x}^c, r_{i,y}^c, r_{i,z}^c)^\top. \tag{5}$$

Let h_x be the half-thickness of the cutting head along the feed axis, h_y the half-width of the blade, ℓ_{comb} the forward

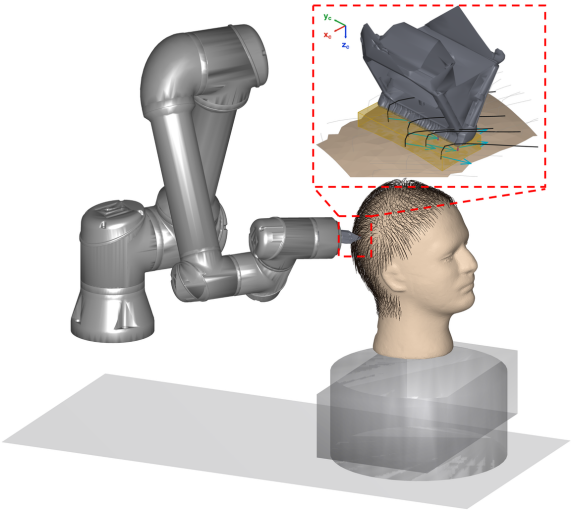


Fig. 3. Geometry of the effective-cutting criterion and the scene constraints at one executed cutting pose. The overview shows the robot arm, the clipper, the head with its hair, and the torso, shoulder and floor obstacles. The inset shows the clipper and the scalp in the TCP frame. The yellow region is the action range of the clipper, drawn only above the scalp, and the cyan arrows are the hair flow directions $\boldsymbol{f}_i$ at the roots. Purple hairs have their root node in the action range and are pushed against the flow, so they are cut to the target node (orange). Black hairs have their root node in the action range but violate the direction condition and are not cut, while gray hairs lie outside the action range of the clipper.

extension of the comb, h_0 the nominal standoff, and τ_h and τ_l the upper and lower height tolerances, where τ_l admits root nodes slightly above the teeth face because of the root offset and the scalp curvature. The geometric action criterion is

$$G_i(T_{wc}) = \mathbf{1}\Big[-h_x \le r^c_{i,x} \le h_x + \ell_{\mathrm{comb}}, \quad |r^c_{i,y}| \le h_y, \\ -\tau_l \le r^c_{i,z} \le h_0 + \tau_h \Big], \tag{6}$$

where $\mathbf{1}[\cdot]$ equals 1 when all conditions hold and 0 otherwise. With the feed direction $\hat{\boldsymbol{x}}_c = R_{wc}(1,0,0)^\top$ and a maximum against-flow angle $\gamma_{\max}$, the effective-cutting criterion, illustrated in Fig. 3, is

$$E_i(T_{wc}) = G_i(T_{wc})\, \mathbf{1}\left[\hat{\boldsymbol{x}}_c^\top(-\boldsymbol{f}_i) \ge \cos\gamma_{\max}\right]. \tag{7}$$

Only cutting-phase poses take part in cutting. Consecutive cutting poses are densified by linear interpolation of position and spherical interpolation of rotation, with steps not exceeding Δ_{tr} and Δ_{rot}, and with $\mathcal{P}_t$ the densified poses of one update, the cutting state changes as

$$a_i(t+1) = \begin{cases} a_i^\star, & i \in \mathcal{U}(t),\ \exists T \in \mathcal{P}_t : E_i(T) = 1, \\ a_i(t), & \text{otherwise.} \end{cases} \tag{8}$$

5) Scene constraints: The scene constraints decide whether a clipper pose or an arm configuration is admissible, independently of the cutting criterion. The links of the arm must keep a safety margin m_{rob} from the obstacle set $\mathcal{O}$, which contains the convex hull of the head, $\mathrm{conv}(\mathcal{M})$, and, for a floor-mounted base, a torso cylinder, a shoulder box and a floor slab, denoted $\mathcal{O}_{\mathrm{env}}$. With $\mathcal{L}_l(\boldsymbol{q}; T_{wb})$ the space occupied by link l in configuration $\boldsymbol{q}$ and base pose T_{wb}, the constraint on the links is

$$\mathrm{dist}\left(\mathcal{L}_l(\boldsymbol{q}; T_{wb}),\, \mathcal{O}\right) \ge m_{\mathrm{rob}}, \qquad l = 1, \ldots, n_q, \tag{9}$$

and nonadjacent links and the clipper must not collide with each other. The convex hull bridges the concavities of the head at the nape, behind the ears and along the face, where a clipper close to the scalp already lies inside the hull. The clipper is therefore checked against the head mesh itself, and its clearance, the minimum signed distance from its points to the mesh, must satisfy

$$d_c(T_{wc}) = \min_{\boldsymbol{x} \in \mathcal{Q}(T_{wc})} d_{\mathcal{M}}(\boldsymbol{x}) \ge \delta_c, \tag{10}$$

where δ_c is the minimum clearance.

III. Planning, Teleoperation and Evaluation

Building on the scene model, this section first plans the ideal toolpath and then the motion of the robot, describes how a person can drive the robot by teleoperation, and finally defines the coverage evaluation of each stage.

A. Task Planning

Task planning works region by region. For a region Ω, it targets the hairs of $\mathcal{U}(t)$ whose roots lie in Ω and outputs an ordered set of toolpath segments, each a sequence of TCP poses labeled as approach, cutting or retraction. Since planning involves only the head, the hair and the clipper, the same ideal toolpath can be handed to different mounting configurations. The reference method, hair-flow-aware parallel toolpath planning, combines a generic parallel toolpath procedure with an against-flow strategy and residual cuts.

1) Generic toolpath generation: Toolpaths are planned on a local machining surface made of the triangles near the roots of the region. This surface extends into the neighboring regions by an overlap band, so toolpaths can cross region boundaries and also cut the neighboring hairs they pass over.

a) Parallel sections: For a group of roots g with path direction $\hat{\boldsymbol{d}}_g$ in the tangent plane of their mean normal $\overline{\boldsymbol{n}}$, a family of parallel planes with normal $\boldsymbol{\zeta} = \overline{\boldsymbol{n}} \times \hat{\boldsymbol{d}}_g$ is intersected with the machining surface. This gives scalp polylines along $\hat{\boldsymbol{d}}_g$, which are trimmed to the roots of the group and resampled by arc length. Adjacent toolpaths are spaced by $\Delta_w = 2h_y(1-\eta)$, set by the blade width and the overlap ratio η, and the plane offset is reduced where the scalp is inclined to the planes. Each polyline is then cut into presegments of length ℓ_{seg}, and with $\ell_{\mathrm{seg}} = \infty$ a toolpath forms a single segment.

b) Tool poses: The k-th scalp point $\boldsymbol{p}^s_k$ of a segment gives the nominal TCP pose T^0_k of the centered blade, located at $\boldsymbol{p}^s_k + h_0 \boldsymbol{n}_k$ with the smoothed outward normal $\boldsymbol{n}_k$. Its z axis is $-\boldsymbol{n}_k$, and its x axis is the polyline tangent projected onto the tangent plane. Where the centered blade cannot reach, the blade can also be offset along its width and rolled about the feed axis to cut with its edge. A pose that violates the clearance condition of Eq. (10) is raised along $\boldsymbol{n}_k$ by at most τ_h, the height tolerance of Eq. (6), so the lifted pose stays within the standoff range admitted by the cutting criterion. If

TABLE II
MODELING DIFFERENCES BETWEEN THE THREE MOUNTING CONFIGURATIONS.

	Fixed rear	Ceiling	Rail
Base location	Rail station behind the head	Above the head center, inverted	Discrete rail stations
Key parameters	Rail radius r_{rail}, station θ_0	Mounting height h_c	Rail radius r_{rail}, station set Θ
Decision variables	$\boldsymbol{q}$	$\boldsymbol{q}$	$\boldsymbol{q}$, segment station θ_j
Obstacle set	$\mathrm{conv}(\mathcal{M}) \cup \mathcal{O}_{\mathrm{env}}$	$\mathrm{conv}(\mathcal{M})$	$\mathrm{conv}(\mathcal{M}) \cup \mathcal{O}_{\mathrm{env}}$
Segment not executable	Feasibility splitting	Feasibility splitting	Change station, then feasibility splitting
Transit	Joint space	Joint space	Joint space, with rail moves between stations via the parking configuration

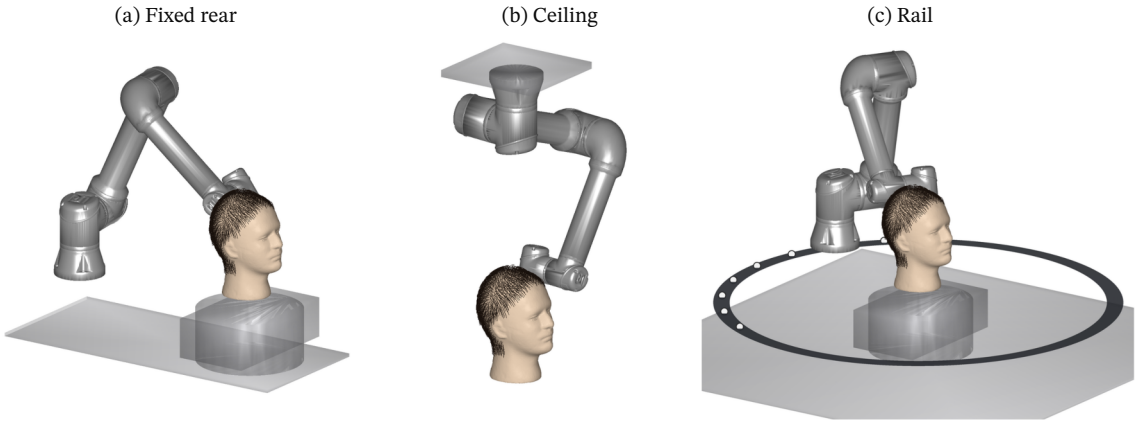


Fig. 4. The three mounting configurations, each at one executed cutting pose of the same scalp region. (a) The fixed rear configuration has its base at the station θ_0 behind the head, (b) the ceiling configuration mounts its base upside down above the head, and (c) the base of the rail configuration moves between the discrete stations Θ (white dots) on a circular rail of radius r_{rail}. The floor, torso and shoulders are drawn semitransparent, and the convex hull of the head is not drawn.

no lift up to τ_h is sufficient, the segment is split at that point. Approach and retraction motions along the scalp normal are finally added at both segment ends.

2) Against-flow strategy: By Eq. (7), the guard comb lifts and cuts a hair only when the clipper is pushed against its flow. Toolpaths that follow the region shape alone, such as common back-and-forth rasters, partly run with or across the flow, so the against-flow strategy takes the path direction from the hair flow and pushes the clipper against it. Near the crown, however, the hair splits forward and backward at the highest point of the head, and one path direction cannot oppose both groups. The roots of a region are therefore grouped by their direction coherence $\rho = \|\frac{1}{N_r}\sum_i \boldsymbol{f}_i\|$ over the N_r roots, which is close to 1 for similar flow directions and small when opposite directions cancel. When ρ is below the threshold $\rho_{\min}$, the roots are split into two groups along the principal axis of their flow directions, and otherwise the region forms one group. Each group takes the projection of its mean flow direction as path direction $\hat{\boldsymbol{d}}_g$, and each toolpath is fed against the mean flow of the nearby roots.

The main passes inevitably miss a few roots, and those not cut effectively form the residual set $\mathcal{U}_r$. Residual cuts take one cluster of neighboring residual roots at a time, generate candidate short toolpaths around it with different feed directions and blade poses, and add the candidate that cuts the most roots of $\mathcal{U}_r$. This repeats until $\mathcal{U}_r$ is empty or an iteration limit is reached, and the main passes and residual toolpaths together form the ideal toolpath of the region.

B. Robot Motion Planning

Robot motion planning turns the ideal toolpath into a joint trajectory of an n_q-degree-of-freedom arm. It decides which segments the arm can execute, giving the executable segments $\mathcal{J}_{\mathrm{exec}}$ and the coverage C_{exec}, and which of them can be joined into one continuous motion, giving the execution chain $\mathcal{J}_{\mathrm{chain}} \subseteq \mathcal{J}_{\mathrm{exec}}$ and the coverage C_{chain}.

1) Mounting configurations: Three mounting configurations share the ideal toolpath, inverse kinematics, scene constraints and transits, and differ only in the base location, the obstacles and whether the base can move (Table II, Fig. 4). The fixed rear and rail bases stand on a circular floor rail of radius r_{rail} around the head with admissible stations Θ. The fixed rear configuration uses only the station θ_0 behind the head, while the rail configuration chooses a station $\theta_j \in \Theta$ for each segment. The ceiling configuration mounts the base upside down at height h_c above the head.

2) Segment executability: A segment is solved sample by sample at a station, from approach through cutting to retraction. For each sample, numerical inverse kinematics from several seeds, starting with the previous configuration $\boldsymbol{q}_{k-1}$, minimizes the weighted flange pose error with the position weighted far above the orientation, giving candidates $\widehat{\boldsymbol{q}}^{(n)}$ that are ranked by

$$\begin{aligned}\Phi(\boldsymbol{q}) = \|\boldsymbol{p}_c(\boldsymbol{q}) - \boldsymbol{p}_{c,k}\| + \lambda_q\|\boldsymbol{q} - \boldsymbol{q}_{k-1}\| \\ + \lambda_R\,\vartheta_R(\boldsymbol{q}) + \lambda_a\vartheta_a(\boldsymbol{q}) + \lambda_f\vartheta_f(\boldsymbol{q}),\end{aligned} \tag{11}$$

where $\boldsymbol{p}_c$ and $\boldsymbol{p}_{c,k}$ are the realized and target TCP positions, ϑ_R is the TCP rotation error, ϑ_a and ϑ_f are the errors of the blade normal and the feed direction, and large λ_a, λ_f keep the two directions that decide cutting close to their targets. A candidate is feasible if it reaches the target position within a tolerance ($\mathcal{Q}_{\mathrm{IK}}$), satisfies Eq. (9) without self-collision ($\mathcal{Q}_{\mathrm{col}}$), changes no joint by more than a maximum step from $\boldsymbol{q}_{k-1}$, which prevents jumps between solution branches ($\mathcal{Q}_{\mathrm{step}}$), and keeps the clipper clearance of Eq. (10) ($\mathcal{Q}_{\mathrm{clr}}$). The feasible candidates of sample k thus form the set

$$\mathcal{Y}_k = \{\widehat{\boldsymbol{q}}^{(n)}\}_n \cap \mathcal{Q}_{\mathrm{IK}} \cap \mathcal{Q}_{\mathrm{col}} \cap \mathcal{Q}_{\mathrm{step}} \cap \mathcal{Q}_{\mathrm{clr}}. \tag{12}$$

The sample takes $\boldsymbol{q}_k = \arg\min_{\mathcal{Y}_k} \Phi$, and the realized TCP pose of this configuration

$$\widehat{T}_{wc,k} = \mathrm{FK}(\boldsymbol{q}_k; T_{wb})\, T_{ec} \tag{13}$$

is used for cutting and evaluation, so a pose deviation matters only if it changes the result of Eq. (7). If $\mathcal{Y}_k$ is empty for any sample, the segment is not executable at that station. The rail configuration then tries its other stations, and if none succeeds, *feasibility splitting* bisects the cutting interval recursively, up to depth D_{split}, and solves the parts separately, so that a short unreachable piece does not discard the whole segment.

3) Transits and execution chain: A transit between two segments retreats the clipper from the head with fixed orientation, transfers along a straight joint-space line or, if that fails, a spherical shell path around the head, and approaches the next segment. Every sample of a transit must satisfy Eqs. (9) and (10) and the joint step limit. Between different rail stations the arm moves to a parking configuration, and the rail moves while the arm is fixed. The executable segments are joined greedily in planning order into a chain that starts from and returns to the parking configuration, trying a direct transit, a transit via the parking configuration and, for the rail, a cross-station route. Segments that cannot be joined are deferred and retried at the chain end, at other stations or between adjacent chain segments. In a full head haircut the regions are executed in turn on a shared hair state, so hairs already cut in one region leave $\mathcal{U}(t)$ of the later ones.

C. Teleoperation

A planner cuts the hair in the way it was designed to, whereas a barber or caregiver may want to cut it in their own way, for example by choosing the path and the order of the regions, giving regions different lengths, shaping edges with the blade edge or finishing what the robot leaves, such as patches the arm cannot reach or connect (Section IV-B3). RoboHairSim therefore also lets a person drive the robot by keyboard in the same scene. The clipper is steered over the scalp rather than through free space. Commands slide, turn, lift and roll it and, on the rail, move the base, one key turns the feed against the local hair flow, and it cuts only while pushed forward with the cut key held. Every command passes the same pose construction, clearance adjustment and feasibility check as a planned toolpath (Section III-A1, Eqs. (11)–(12)), and a command that fails is rejected with its reason shown. Cutting follows Eqs. (7) and (8), so a hand-guided pass cuts exactly what the same planned pass would. The operator can confine the clipper to one region and set that region's own target length, and the display shows the standoff, the clearance, the angle between the feed and the hair flow and the roots still to be cut. A barber's own cutting plan can thus be tried on the simulated robot under the same safety constraints before any real hair is cut.

D. Coverage Evaluation

Four coverage measures evaluate the haircutting task layer by layer. The first two evaluate the ideal toolpath itself, and the last two evaluate whether the arm can realize it and execute it continuously. A *regional task* starts from the initial hair state $\mathcal{H}^0$ and plans and executes one scalp region, and it is used to compare toolpaths and mounting configurations. A *full head haircut* completes all regions in turn on a shared hair state. With $\mathcal{U}_0$ the set still to be cut at the start of a task and $N_0 = |\mathcal{U}_0|$, the coverage of a set of TCP poses $\mathcal{P}$ under an action criterion $A_i \in \{G_i, E_i\}$ is the fraction of the roots of $\mathcal{U}_0$ that $\mathcal{P}$ acts on,

$$C(\mathcal{P}, A) = \frac{1}{N_0} \sum_{i \in \mathcal{U}_0} \mathbf{1}\big[\exists\, T \in \mathcal{P} :\ A_i(T) = 1\big]. \tag{14}$$

The four measures differ only in the pose set and the criterion.

- The geometric coverage $C_{\mathrm{geom}} = C(\mathcal{P}_{\mathrm{task}}, G)$ uses the cutting poses $\mathcal{P}_{\mathrm{task}}$ of the ideal toolpath and only asks whether the clipper sweeps over a root.
- The task coverage $C_{\mathrm{task}} = C(\mathcal{P}_{\mathrm{task}}, E)$ uses the same poses but requires effective cutting, so $C_{\mathrm{task}} \le C_{\mathrm{geom}}$.
- The single segment coverage $C_{\mathrm{exec}} = C(\widehat{\mathcal{P}}_{\mathrm{exec}}, E)$ uses the realized TCP poses (Eq. (13)) of the executable segments $\mathcal{J}_{\mathrm{exec}}$, whether or not they can be connected.
- The execution chain coverage $C_{\mathrm{chain}} = C(\widehat{\mathcal{P}}_{\mathrm{chain}}, E)$ uses only the realized poses of the execution chain $\mathcal{J}_{\mathrm{chain}}$. It measures what the arm achieves when it starts from the parking configuration, executes continuously and returns, and it is the main measure, with $C_{\mathrm{chain}} \le C_{\mathrm{exec}}$.

The difference between adjacent layers is the loss attributed to that layer. $C_{\mathrm{geom}} - C_{\mathrm{task}}$ is the task layer loss caused by the direction condition, $C_{\mathrm{task}} - C_{\mathrm{exec}}$ the realization loss from segments that cannot be executed, and $C_{\mathrm{exec}} - C_{\mathrm{chain}}$ the connection loss from transits. The last two form the robot layer loss, relative to the motion planning method used. This decomposition holds if the realized poses cut no roots beyond those cut by the ideal toolpath, a premise checked task by task in the experiments. Regional results are aggregated as the mean of the region coverages C^{Ω} weighted by their initial numbers of roots to be cut N_0^{Ω}, while a full head haircut uses the set to be cut of the whole head as the denominator.

IV. Experiments

This section applies RoboHairSim to one haircutting scene. It first compares seven combinations of the toolpath components at the task layer, then analyzes the robot layer loss of the hair-flow-aware toolpaths under three mounting configurations and two execution modes, and finally performs continuous full head haircuts and a teleoperated haircut of one region.

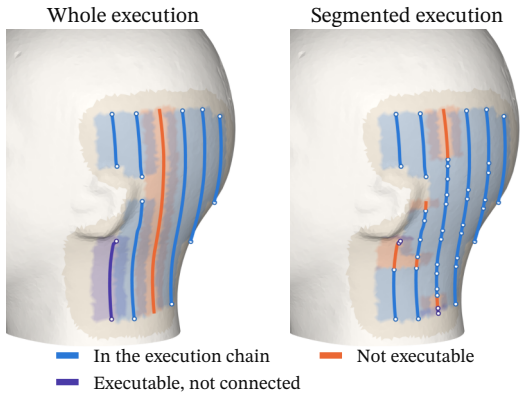


Fig. 5. Whole pass (left) and segmented (right) execution of the left region with the fixed rear configuration (hair-flow-aware parallel toolpaths, 5 mm). Blue segments are in the execution chain, purple segments are executable but could not be connected by transits, and orange toolpaths are not executable. The translucent bands of the same colors show the scalp swept by the blade along these toolpaths, and circles mark segment ends.

A. Experimental Setup

All experiments use the same head mesh and procedural hair model. The top left, top right, left, right and back regions contain 579, 589, 303, 319 and 410 hairs, with initial lengths of 18.4–35.9 mm (mean 27.4 mm). The regional tasks use target lengths of 5 and 20 mm. At 5 mm all 2200 hairs must be cut, whereas at 20 mm part of the hair on the sides is already short enough and 1980 hairs remain to be cut, and the full head haircuts also use 0 mm. Each regional task starts from the initial hair state, and the results of the five regions are aggregated as in Section III-D.

The task layer experiments evaluate only the ideal toolpaths of the five regions, without the robot. The robot layer experiments use the hair-flow-aware parallel toolpaths of Section III-A, hereafter the flow-aware method, on a UR5e in the fixed rear, ceiling and rail configurations, each with two execution modes. In whole pass execution, each parallel toolpath forms a single segment without feasibility splitting, so it is discarded entirely if any of its samples cannot be executed. In segmented execution, the toolpaths are presegmented to a fixed length and a segment that cannot be executed is split as in Section III-B2 (Fig. 5). The five regions at both target lengths thus give 60 regional tasks. All experiments run in MATLAB R2025b on an Apple M2 computer, and computation times are measured single-threaded.

B. Coverage Analysis

1) Target length: The target length decides which hairs must be cut and how much is removed from each (Fig. 6). The initial hair is longer on the top and back than on the sides, so the mean removed length drops from 21.2 mm at 5 mm to 6.2 mm at 20 mm, when about one third of the hairs on the sides need no cut. Because the effective-cutting criterion does not depend on the target length, the toolpaths and robot trajectories are the same at both lengths, and the execution chain coverage differs by at most 2.0 percentage points. The results at 5 mm are therefore used in the rest of this section.

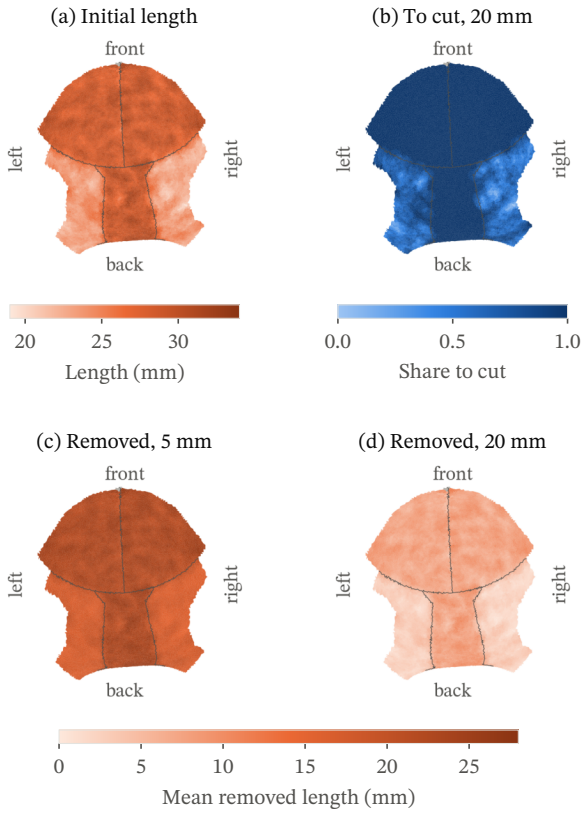


Fig. 6. Effect of the target length on the task (unwrapped scalp). (a) Initial hair length, (b) share of roots to be cut at 20 mm (all roots are cut at 5 mm), and (c, d) mean removed length at 5 and 20 mm. Each panel is an azimuthal equidistant projection about the center of a sphere fitted to the scalp, with the frontal hairline at the top, the nape at the bottom and left and right as seen by the subject. Thin lines mark the borders of the five scalp regions, and each value is the mean over the roots within a radius of 10 mm.

2) Task layer: The flow-aware toolpath is determined by four components, namely the path direction, the grouping by hair flow, the feed direction and the residual cuts. Table III compares seven combinations of them, built up step by step from the geometric raster common in generic surface coverage.

- V1 is the geometric raster, with toolpaths along the longest direction of the roots in the mean tangent plane, alternating feed, and no grouping or residual cuts.
- V2 feeds every toolpath of V1 against the mean flow of the hair near it.
- V3 groups the roots of V2 by hair flow, each group keeping its geometric axis. V4 adds residual cuts to V3.
- V5 aligns the toolpaths of V2 with the mean hair flow of the region, without grouping.
- V6 groups the roots of V5 so that the toolpaths of each group follow its own hair flow, and V7, the flow-aware method, adds residual cuts.

A toolpath with complete geometric coverage can have a much lower task coverage. All combinations sweep over at least 99.9% of the roots to be cut, yet the geometric raster (V1) moves with or across the hair flow at about 29% of them and reaches a task coverage of only 71.05%, while the flow-aware method (V7) reaches 100%.

TABLE III
TASK COVERAGE OF THE TOOLPATH COMBINATIONS.

					Top regions				Sides and back	All
	Path direction	Grouping	Feed	Residual cuts	C_{main}	C_{task}	Residual segments	Toolpath length (m)	C_{task}	C_{task}
V1	Geometric axis	No	Back and forth	No	68.92	68.92	0	2.82	73.45	71.05
V2	Geometric axis	No	Against flow	No	57.62	57.62	0	2.83	99.81	77.41
V3	Geometric axis	Yes	Against flow	No	17.89	17.89	0	3.02	–	–
V4	Geometric axis	Yes	Against flow	Yes	17.89	100.00	71	5.80	–	–
V5	Hair flow	No	Against flow	No	28.51	28.51	0	2.74	100.00	62.05
V6	Hair flow	Yes	Against flow	No	93.49	93.49	0	2.97	100.00	96.55
V7	Hair flow	Yes	Against flow	Yes	93.49	100.00	16	3.39	100.00	100.00

Note. Coverages are in percent at a target length of 5 mm and are weighted by the number of roots to be cut of each region. C_{main} is the task coverage of the main passes alone. The top regions are reported separately from the side and back regions because their hair flow differs. Residual segments and toolpath length are summed over the two top regions, and V3 and V4 are computed on the top regions only.

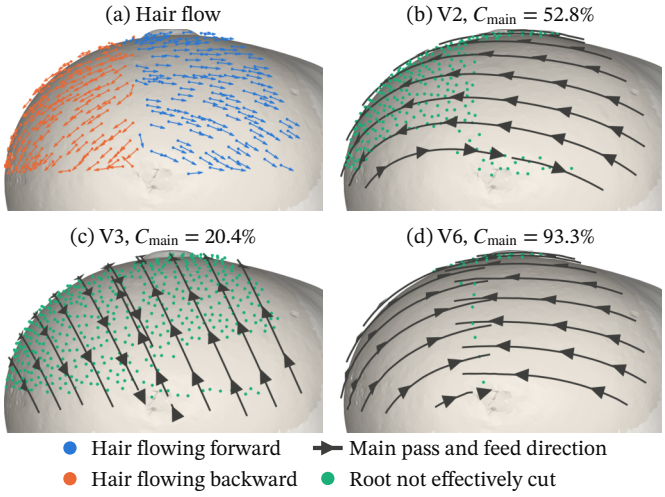


Fig. 7. Main passes of the top left region under three toolpath combinations (view from above, forehead to the right, 5 mm). (a) Hair flow direction, with the groups flowing forward and backward in blue and orange. (b) V2 with the geometric axis, against-flow feed and no grouping. (c) V3 with the geometric axis, against-flow feed and grouping. (d) V6 with the hair flow, against-flow feed and grouping. Dark gray lines are the main passes with arrows for the feed direction, green dots are roots to be cut that the main passes do not cut effectively, and each title gives the main pass coverage C_{main} of the region.

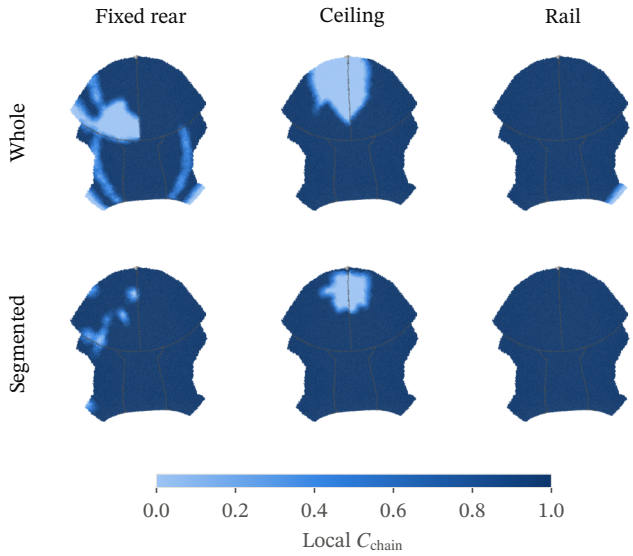


Fig. 8. Local execution chain coverage of the flow-aware method (5 mm, unwrapped scalp with the frontal hairline at the top). The rows show whole pass and segmented execution, and the columns the fixed rear, ceiling and rail configurations. Each value is the share of the roots to be cut within a radius of 10 mm of the point that are covered by the execution chain.

On the side and back regions the hair flow is uniform, so the direction coherence ρ stays above $\rho_{\min}$ and no combination groups the roots. Pushing every toolpath against the nearby hair (V2) already makes the task coverage of these regions nearly complete, whether the toolpaths follow the geometric axis or the hair flow (V5).

On the top regions the hair splits forward and backward at the highest point of the head (Fig. 7a). Without grouping, a toolpath crossing the region can oppose only one of the two groups (V2, Fig. 7b), and toolpaths along the mean flow cover even less (V5), since this mean deviates from both groups. After grouping, the geometric axis of each group turns sideways and the toolpaths run nearly perpendicular to the flow, which gives the lowest main pass coverage (V3, Fig. 7c). With grouping and toolpaths along the flow of each group, the main pass coverage reaches 93.49% (V6, Fig. 7d). The effect of grouping thus depends on the path direction, lowering the main pass coverage with the geometric axis and raising it sharply with the hair flow.

Residual cuts bring every combination to 100%, and their amount follows the main passes. On the top regions V4 needs about 4.4 times as many residual segments as the flow-aware method, and its toolpaths are 71% longer. At 20 mm the ranking of the five-region task coverage is unchanged. With a maximum against-flow angle $\gamma_{\max}$ of 30° the flow-aware method still reaches 99.27%, and it reaches 100% from 45° to 90°.

3) Robot layer: Since the task coverage of the flow-aware method is 100%, any shortfall of the execution chain coverage comes from the robot layer (Table IV), and all 60 regional tasks satisfy the premise of the loss decomposition. In whole pass execution, a toolpath with a locally unreachable part is discarded entirely, and the robot layer loss of the fixed rear and ceiling configurations is mainly realization loss. Segmented execution splits such toolpaths and raises the execution chain

TABLE IV
COVERAGE, ROBOT LAYER LOSS AND COMPUTATION TIME OF THE FLOW-AWARE METHOD FOR EACH MOUNTING CONFIGURATION.

Configuration	Execution	C_{exec}	C_{chain}	Realiz. loss	Conn. loss	C_{chain} (20 mm)	Motion planning (s)	Transit share (%)	Transits (feasible/tried)
Fixed rear	Whole pass	85.82	81.32	14.18	4.50	81.36	147	51	58/116
Fixed rear	Segmented	98.91	96.64	1.09	2.27	96.46	381	62	183/349
Ceiling	Whole pass	84.14	81.82	15.86	2.32	79.80	118	50	49/64
Ceiling	Segmented	91.59	91.45	8.41	0.14	90.51	216	42	153/168
Rail	Whole pass	98.95	98.95	1.05	0.00	99.19	261	55	68/72
Rail	Segmented	100.00	100.00	0.00	0.00	100.00	392	57	157/163

Note. Coverages and losses are in percent at 5 mm, except the column C_{chain} (20 mm). The realization loss is $C_{\text{task}} - C_{\text{exec}}$ and the connection loss $C_{\text{exec}} - C_{\text{chain}}$, and $C_{\text{geom}} = C_{\text{task}} = 100$ in every configuration. The motion planning time is summed over the five regional tasks at 5 mm and includes infeasible transit trials, while task planning takes 130–134 s in every configuration.

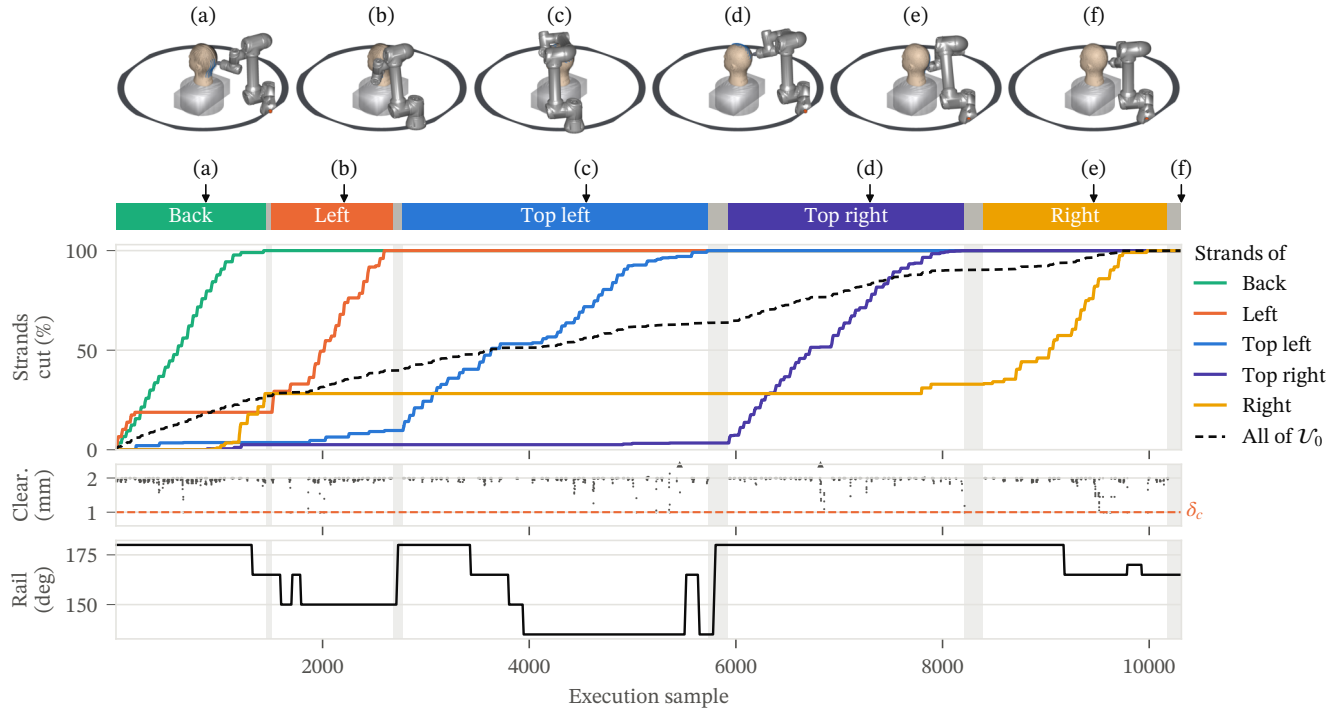


Fig. 9. Course of the continuous full head haircut (flow-aware method, rail configuration, segmented execution, 5 mm). The top row shows the scene midway through each region (a)–(e) and at the end of the haircut (f), with the toolpaths already executed in the current region in blue. The color strip below gives the region of each execution sample, with transits in gray, and arrows mark the samples shown above. The three plots show the share of the hairs of each region cut to their target node, with the whole $\mathcal{U}_0$ dashed, the clipper clearance on cutting samples, with the minimum clearance δ_c dashed and samples beyond the axis drawn on the top edge, and the rail station angle, with transits as light gray bands.

coverage in all three configurations, most with the fixed rear configuration. Connection loss then dominates with the fixed rear configuration, realization loss still dominates with the ceiling configuration, while the rail configuration shows no robot layer loss at all.

Fig. 8 locates these losses on the scalp. In whole pass execution, the fixed rear configuration loses coverage at the junction of the top left and left regions and along the borders of the back region, the ceiling configuration on both sides of the midline at the front, and the rail configuration only in a small patch near the nape. With segmented execution, only a few small patches remain with the fixed rear configuration and a smaller front area with the ceiling configuration, and none with the rail configuration.

C. Safety and Computation Time

Over all cutting and transit samples of the executed trajectories, the clipper clearance never falls below $\delta_c = 1$ mm and the distance from the links to the obstacles never below 1.06 mm. The realized TCP poses differ from the ideal ones by less than 0.01 mm and $0.01°$, except with the ceiling configuration in segmented execution, where the deviations reach 1.9 mm and $3.96°$. These deviations cause no missed cuts, so the robot layer loss comes entirely from segments that cannot be executed and from failed transits.

Task planning takes 130–134 s, about half of it for residual cuts, while motion planning takes 118–392 s (Table IV). Transits account for 42–62% of the motion planning time, because every transit step solves inverse kinematics and failed transits are retried. Segmented execution, with more segments

and feasibility splitting, takes 1.5–2.6 times as long as whole pass execution.

D. Full Head Haircuts

RoboHairSim also simulates a complete haircut. With the rail configuration and segmented execution, the flow-aware method cuts the five regions in turn on a shared hair state, connects them by transits and rail moves, and returns the arm to the parking configuration at the end. At target lengths of 0, 5 and 20 mm, every region reaches an execution chain coverage of 100%, and all hairs of $\mathcal{U}_0$ end at their target node. The remaining length error stems from the discrete hair nodes, since the retained length slightly exceeds the target. The mean errors are 1.15 and 1.25 mm and the maximum errors 2.50 and 2.85 mm at 5 and 20 mm, and there is no error at 0 mm. The minimum clipper clearance over the three haircuts is 1.00 mm.

Fig. 9 shows the course of the 5 mm haircut. The hairs of each region are mostly cut while that region is executed, but toolpaths at region edges cross the boundaries, so 238 hairs (10.8%) are cut by a neighboring region before their own region is reached. In the right region this holds for 32.9% of the hairs, which are cut while the back and top right regions are executed. The rail switches stations both between and within regions, at angles from 135° to 180°.

E. Teleoperated Haircut

RoboHairSim also supports cutting by teleoperation in the same scene. One operator cut the left region at 5 mm by keyboard with the rail configuration, with the clipper confined to that region (Fig. 10). The operator chose the passes and their order, turned the feed against the hair flow, moved the base along the rail where the arm could not reach, and lifted the clipper between passes. Every command passed the same feasibility check as a planned toolpath, and a rejected command left the robot at its last admissible pose. Evaluated with the same effective-cutting criterion as the planned toolpaths, the realized motion cut all 303 hairs of $\mathcal{U}_0$ in 311 s, a coverage of 100%. Of the 73 rejected commands, 64 commands would have moved the clipper out of the region and 9 commands would have brought an arm link into contact with the head, so both the region confinement and the collision check were active during the haircut.

V. Conclusion

This paper presented RoboHairSim, a simulation platform for planning and evaluating robotic haircutting. It models the haircutting scene, plans clipper toolpaths and robot motion, lets a person drive the same robot by teleoperation, and evaluates planned and executed results with one effective-cutting criterion based on the guard comb. The experiments on one head showed that geometric coverage alone overestimates what a toolpath achieves. Against-flow feed nearly completes the side and back regions, while the crown requires grouping the roots by hair flow and running the toolpaths of each group along its flow. The layered coverage metrics further separated the execution losses of the robot into segments that cannot be executed and failed transits, and located them on the scalp. A teleoperated haircut of one region cut all of its hairs, while the platform rejected the commands that would have left the region or brought the arm into contact with the head.

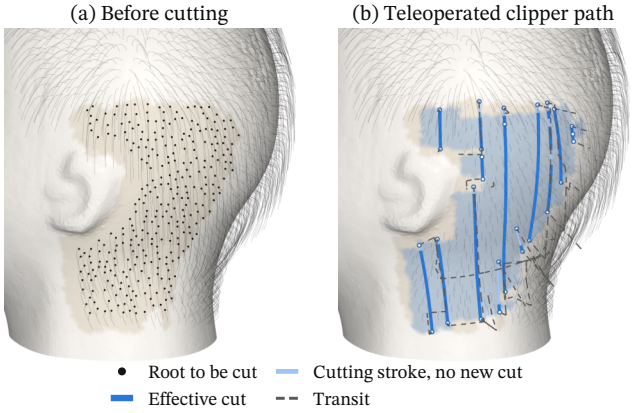


Fig. 10. Teleoperated haircut of the left region (rail configuration). (a) Before cutting, with the roots to be cut. (b) Executed clipper path, in which blue moves cut hair, light blue moves pass over hair already cut and dashed moves are transits, with the blue band showing the scalp swept by the blade.

The platform also has limitations that point to future work. The parameters of the effective-cutting criterion, such as the action range, the height tolerances and the maximum against-flow angle, have not yet been measured, and the contact between the comb, the blade and the hair is not modeled. Only one head and one hairstyle were used, and head motion and perception errors were not considered. Future work will compare simulated and real haircuts on mannequin heads, evaluate the teleoperation mode with barbers and caregivers, study shared control between a planner and a teleoperating person, and use the coverage layers as objectives or rewards for planning methods, including those for graded hairstyles defined by per-hair target lengths.

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